FedPDFGuard: A Federated Learning Approach for Robust Detection of Malicious PDF Files

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***Abstract*—FedPDFGuard is the first federated learning-based system for PDF malware detection. It enables decentralized model training across participants while preserving data privacy. Our evaluation shows that FedPDFGuard achieves detection accuracy comparable to centralized methods, addressing privacy concerns and improving robustness against increasingly sophisticated PDF-based cyberattacks.**

***Index Terms*—PDF malware, intrusion detection, federated learning**

I. Introduction

The Portable Document Format (PDF) has become a standard for digital document sharing due to its platform independence, consistent formatting, and compatibility with a wide range of devices and operating systems. Its ubiquity is further reinforced by the widespread use of PDF readers, including popular tools like Adobe Acrobat Reader and Apple Preview, as well as numerous open-source projects and browser plugins, all installed on millions of devices worldwide. However, the design of the PDF format allows for embedded executable scripts, such as JavaScript snippets, which can be exploited for malicious purposes. Additionally, the complexity of PDF readers often results in vulnerabilities that cybercriminals can target. These factors make malicious PDF files (i.e., PDF malware) an attractive attack vector for exploiting vulnerable PDF readers at scale. For example, PDF malware can execute shellcode to exploit reader vulnerabilities or download malicious binaries from external sources. A 2023 report by Palo Alto Networks revealed that PDFs were the primary type of malicious email attachment, accounting for 66% of such attachments used to deliver malware via email [1]. Moreover, these attacks are rapidly evolving and becoming increasingly sophisticated and stealthy. Tom's Guide recently reported cases of malicious PDF files embedding Word documents, which in turn contain macros designed to download and install malware upon opening [2].

Detecting PDF malware is of critical importance. Existing approaches [3]–[7] predominantly leverage machine learning techniques for detection. Despite variations in model selection and feature sets, they all adhere to a centralized training paradigm, where labeled data are first aggregated into a dataset, and a single detection model is then trained. As PDF malware rapidly evolves, maintaining detection efficacy requires continuously integrating diverse and up-to-date PDF samples into the centralized dataset, ensuring the development of more robust and adaptive detection models.

Nevertheless, continuously obtaining PDF files, particularly malicious samples, from various sources presents significant practical challenges. A primary obstacle is the issue of privacy, as both benign and malicious PDF files may contain sensitive personal or business information. For instance, the publicly available corpus of PDF malware is very limited. To address this gap, we propose a novel system, namely FedPDFGuard, for detecting PDF malware using federated learning [8]. Federated learning (FL) enables collaborative model training across multiple decentralized datasets without requiring data to be shared or centralized, thus preserving privacy. FedPDFGuard leverages this paradigm by training models locally on data from multiple participants, aggregating the insights into a robust global model while ensuring that sensitive information remains secure. We have extensively evaluated FedPDFGuard using a publicly available dataset [5], and our experimental results demonstrate that FedPDFGuard achieves a high detection rate of 99.52% with a very low false positive rate of 1.71%. In summary, our work makes the following key contributions:

- To the best of our knowledge, FedPDFGuard is the first PDF malware detection system built on a federated learning framework to ensure high detection accuracy while preserving the privacy of local clients' data.
- We evaluate the global model accuracy, detection rate, and false positive rate of FedPDFGuard, comparing its performance with three centralized machine learning approaches. Experimental results show that FedPDFGuard achieves comparable effectiveness to centralized methods.
- The source code for FedPDFGuard is publicly available at https://github.com/jzhang369/FedPDFGuard to support reproducibility and encourage future extensions.

The remainder of the paper is organized as follows: Section II reviews the related work. Section III introduces the background of the PDF document structure and malicious PDF files. Section IV details the design of FedPDFGuard, Section V presents the evaluation results. The extensibility of FedPDFGuard is discussed in Section VI, and conclusions are provided in Section VII.

Prototype: Simulated Malicious PDF

```
%PDF-1.5
1 0 obj
<<
  /Type /Catalog
  /OpenAction 2 0 R
>>
endobj

2 0 obj
<<
  /Type /Action
  /S /JavaScript
  /JS (app.alert("Alert!");)
>>
endobj

xref
0 3
0000000000 65535 f
0000000010 00000 n
0000000070 00000 n
trailer
<< /Root 1 0 R /Size 3 >>
startxref
143
%%EOF
```

Fig. 1: A prototype example of a PDF embedding JavaScript

II. Related Work

Several methods have been proposed for detecting PDF malware [3]–[7], with most leveraging machine learning techniques, similar to many intrusion detection systems. Some approaches focus on exploring different machine learning models for detection. For example, Al-Haija et al. [3] employ decision trees, while Andrew et al. [4] utilize image processing models. Issakhani et al. [5] apply K-means clustering to categorize PDF files, labeling those in the "wrong" cluster as malicious. Hidost et al. [6] introduce a feature extraction method based on the parsed tree structure of PDF files, combining these features with decision trees or SVM models for detection. Building on this, Chen et al. [7] assess the robustness of these features and integrate the most resilient ones into neural network models for enhanced detection. While these methods vary in their choice of models and feature sets, they all follow a centralized learning paradigm, where a single detection model is trained on a well-labeled, centralized dataset. In contrast, `FedPDFGuard` eliminates the need for such a centralized dataset. Instead, local models are first trained on individual local datasets and subsequently aggregated into a global model, ensuring a more decentralized and adaptable detection framework.

Federated learning has emerged as a promising approach for intrusion detection across various domains, particularly those where data privacy is a primary concern. For instance, Li et al. [9] leverage federated learning to detect DDoS attacks in Industrial IoT networks, while Filho et al. [10] apply a similar method for botnet traffic detection. Rey et al. [11] introduce a federated learning framework to detect malware in IoT devices, and Zhao et al. [12] develop a federated-learning-based system for identifying network anomalies. Additionally, federated learning has been extensively employed in network traffic classification [13]–[16], analyzing various types of network data such as flows and packets. These studies demonstrate that federated models can achieve performance comparable to centralized models while keeping data localized, thereby preserving privacy. Despite these advancements, there is a notable lack of research on applying federated learning to PDF malware detection. Given the privacy concerns associated with sharing potentially sensitive document data, federated learning offers a compelling solution for developing robust PDF malware detection models without compromising data confidentiality.

III. Background

PDF is a widely adopted standard for document exchange, designed to preserve document formatting across platforms and devices. Defined under ISO 32000-1:2008 [17], the PDF specification outlines a rich and flexible syntax that supports complex features such as text, images, vector graphics, interactive elements, and embedded scripts. To support these capabilities, PDF files are organized into a tree-structured hierarchy composed of objects, streams, and metadata.

Dynamic behaviors in PDF documents are enabled through features such as annotations, actions, and embedded JavaScript. For instance, JavaScript code can be embedded using the `/JavaScript` action and triggered automatically via the `/OpenAction` directive or in response to user interactions such as clicking buttons or filling in form fields. While such scripting capabilities enhance interactivity, they also introduce significant security risks. Attackers can embed JavaScript that executes automatically when the document is opened, exploiting vulnerabilities in the PDF viewer to perform malicious operations. These may include manipulating the document object model (DOM), accessing local system resources, or triggering memory corruption bugs.

To demonstrate how executable logic can be embedded within a PDF, Figure 1 presents a prototype of a simulated malicious file. This example uses the `/JavaScript` action in combination with the `/OpenAction` directive to trigger a script upon document opening. The file begins with a catalog object referencing an action object, which defines a JavaScript payload using the `/JS` field. In this benign prototype, the script simply displays an alert box. However, this same structure can be weaponized by replacing the alert with obfuscated code that exploits vulnerabilities in the PDF reader or downloads and executes remote payloads.

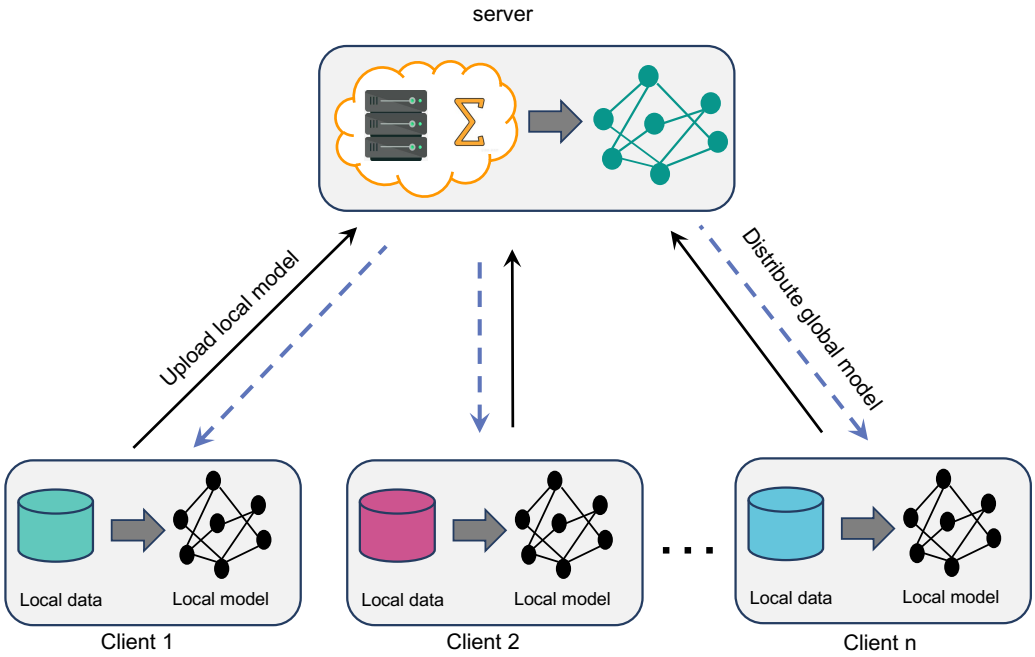


Fig. 2: The Overview of Federated Learning

In real-world attacks, adversaries frequently apply advanced obfuscation techniques—such as hexadecimal encoding, object stream compression, and deeply nested object hierarchies—to evade static analysis and signature-based detection. Embedded scripts may interact with system APIs, modify document content dynamically, or exploit zero-day vulnerabilities in rendering engines. These evasive techniques significantly reduce the effectiveness of conventional detection methods that rely solely on content signatures.

To address these challenges, `FedPDFGuard`analyzes structural features derived from the tree-based hierarchical organization of PDF files. By modeling the document's internal structure rather than its raw content, `FedPDFGuard` offers a robust and privacy-preserving approach to detecting malicious behavior in PDF documents.

IV. Problem Formulation

In a standard federated learning (FL) setting, illustrated in Figure 2, we assume the presence of a central server and n participating clients, denoted as $\mathcal{C} = C_1, C_2, \ldots, C_i, \ldots, C_n$. Each client C_i possesses a local training dataset D_i, where $i = 1, 2, \ldots, n$. The union of all local datasets forms the global training dataset, expressed as $\mathcal{D} = \bigcup_{i=1}^{n} D_i$.

Let $f(w, x_{i,j}, y_{i,j})$ represent the loss function evaluated on the j-th data record of client C_i, where w is the model parameter vector. For notational simplicity, we denote this as $f_{i,j}(w)$. The local loss function at client C_i is then defined as:

$$F_{c_i}(w) = \frac{1}{|D_i|} \sum_{j=1}^{|D_i|} f_{i,j}(w) \tag{1}$$

The loss function at the central server can be represented as a weighted average of all local loss functions, given by:

$$F(w) = \frac{1}{|\mathcal{D}|} \sum_{i=1}^{n} |D_i| F_{c_i}(w) \tag{2}$$

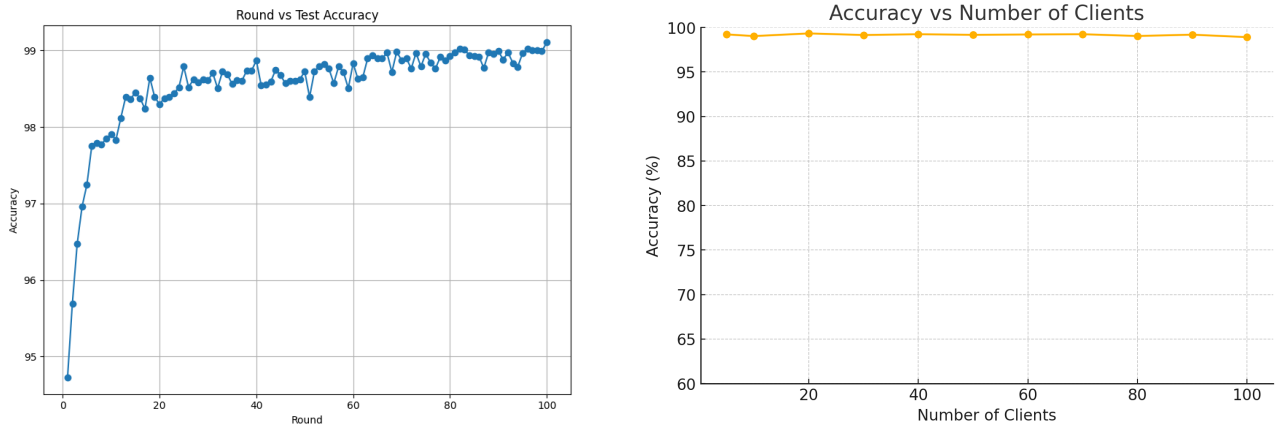
Initially, a global model $w^{(0)}$ is created and randomly initialized, serving as the starting point for the FL process. Each client $C_i \in \mathcal{C}$ then trains a local model $w_i^{(1)}$ using its own dataset D_i. After local training is completed, each client transmits its updated model $w_i^{(1)}$ to the central server. The server aggregates the received local models to produce an updated global model $w^{(1)}$. Following aggregation, the updated global model is distributed back to all clients to begin the next round of training. To illustrate the iterative learning process, we describe the operations performed during the t-th training round as follows:

- **Step 1:** Each client performs local training using gradient descent to update its model based on the global model $w^{(t-1)}$, with a learning rate η_t in the t-th round. The update rule is defined as:

$$w_i^{(t)} = w^{(t-1)} - \eta_t \nabla_{w^{(t-1)}} F_{c_i}(w^{(t-1)}) \tag{3}$$

- **Step 2:** Each client sends its updated local model $w_i^{(t)}$ to the connected edge server.
- **Step 3:** The server aggregates the received local models using a weighted average based on the size of each client's dataset:

$$w^{(t)} = \frac{1}{|\mathcal{D}|} \sum_{i=1}^{n} |D_i| w_i^{(t)} \tag{4}$$



(a) Test accuracy of the global model across training rounds

(b) Accuracy with increasing number of participating clients

Fig. 3: Evaluation of global model performance

- **Step 4:** The server broadcasts the updated global model $w^{(t)}$ to all clients to initiate the next round of training.

These steps are repeated iteratively until the global model's accuracy ceases to improve, indicating convergence. Once convergence is detected, the final global model is distributed to all clients for deployment or further use.

V. Evaluation

A. Dataset and Features

We use the PdfRep dataset [18], which consolidates data from four publicly available sources: the Contagio dataset [19], the CIC dataset [20], the VirusShare dataset [21], and the GovDocs dataset [22]. The combined dataset contains 170,477 benign samples and 322,761 malicious ones.

In [18], each PDF file is represented by the tree structure of its internal objects, from which 12 structural features are extracted. These features include: (1) the average number of children per node, (2) the median number of children, (3) the variance of children, (4) the number of leaves in the tree, (5) the total number of nodes, (6) the average node degree, (7) the degree assortativity coefficient, (8) the average shortest path length, (9) the average clustering coefficient, and (10) the graph density.

These features, when used in combination with statistical classifiers trained on a centralized dataset, have demonstrated high detection accuracy, i.e., ranging from 0.9754 to 0.9878 [18]. Building upon this foundation, we adopt the same feature representation in the design of our federated learning-based PDF malware detection system.

B. Experimental Model

In order to evaluate `FedPDFGuard`, we consider a federated learning system with multiple local clients and a single cloud server. Model training is performed on the aforementioned datasets using a convolutional neural network (CNN), consisting of two convolutional layers with ReLU units, are trained with mini-batch Stochastic Gradient Descent (SGD). The cross-entropy loss function is used throughout the training process.

Both benign and malicious samples are evenly and randomly distributed across all clients. Each client trains its local model using 80% of the training dataset and 20% for testing. The hyperparameters are set as follows: For local training on each client, SGD is used with a batch size of 64. The initial learning rate η for both the local and global models is set to 0.0015 and decays exponentially at a rate of 0.98 per epoch.

C. Detection Performance

Figure 3a presents the test accuracy of the global model across multiple communication rounds. The model shows a consistent improvement in accuracy during the initial rounds and converges after approximately 40 rounds, ultimately reaching an accuracy of 98.28%. This result highlights the effectiveness of our federated learning framework in achieving high detection performance without requiring centralized access to raw data.

The number of participating clients plays a critical role in the performance of federated learning. To evaluate its impact, we examine how varying the number of clients influences model accuracy. As the number of clients increases, the amount of training data available to each individual client decreases. Nevertheless, as shown in Figure 3b, `FedPDFGuard` maintains stable and high detection accuracy, demonstrating strong robustness to the distribution of data across clients. This finding suggests that, in practical deployments, increasing client participation introduces greater diversity and volume of training data, which can lead to a more powerful global model.

However, it is important to note that a larger number of clients may also introduce additional communication overhead and increased data heterogeneity, potentially affecting convergence stability. Despite these challenges, our experimental results indicate that the benefits of scaling the number of clients, particularly in terms of improving model robustness.

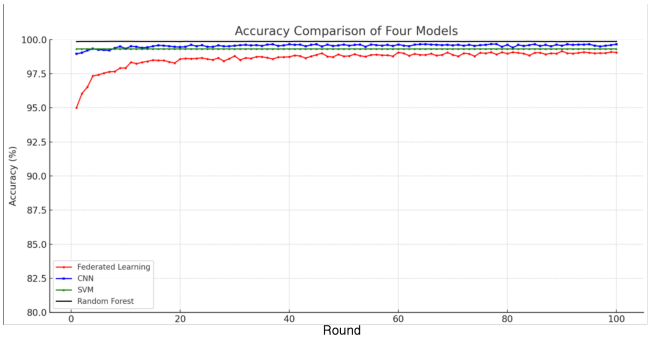


Fig. 4: Accuracy Comparison

D. Accuracy Comparison

To evaluate the effectiveness of FedPDFGuard, we compare its global model accuracy with several widely used centralized machine learning algorithms, including Convolutional Neural Networks (CNN) [23], Support Vector Machines (SVM) [24], and Random Forest [25]. For this comparison, FedPDFGuard is configured with 100 clients and a single central server. To ensure fairness and stability, the reported accuracies for CNN, SVM, and Random Forest are averaged over 100 independent runs.

Figure 4 presents the accuracy comparison among the four models: FedPDFGuard, CNN, SVM, and Random Forest. Random Forest consistently achieves the highest accuracy, maintaining performance close to 99.86% throughout the evaluation. CNN and SVM also perform well, reaching final accuracies of 99.54% and 99.32%, respectively. FedPDFGuard starts with a lower initial accuracy of approximately 95% but steadily improves over time, converging after around 40 rounds at 98.9%. Although FedPDFGuard slightly trails behind the centralized CNN and Random Forest models in terms of final accuracy, it demonstrates strong and competitive performance after sufficient training. These results highlight the capability of FedPDFGuard to achieve high accuracy while maintaining the privacy-preserving advantages of federated learning.

Additionally, Table I provides a detailed comparison of accuracy, detection rate (DR), and false positive rate (FP) for all four models. Random Forest delivers the best overall performance, achieving the highest accuracy (99.86%), the highest detection rate (99.77%), and the lowest false positive rate (0.04%). CNN also performs strongly, with an accuracy of 99.54%, a detection rate of 99.61%, and a low false positive rate of 0.53%. FedPDFGuard attains a competitive accuracy of 98.9% and a high detection rate of 99.52%, although it exhibits a slightly higher false positive rate of 1.71% compared to the centralized models. These results demonstrate that FedPDFGuard provides strong detection performance and competitive accuracy, with the added benefit of enhanced data privacy, at the cost of a modest increase in false positives.

TABLE I: Performance comparison

	Accuracy%	**DR% (Detection Rate)**	**FP% (False Positive)**
FedPDFGuard	**98.9**	**99.52**	**1.71**
CNN	99.54	99.61	0.53
SVM	99.32	99.46	0.82
Random Forest	99.86	99.77	0.04

VI. Discussion

While FedPDFGuard achieves competitive detection performance and demonstrates the feasibility of privacy-preserving PDF malware detection using federated learning, several areas offer potential for enhancement.

1) Incorporating richer semantic and exploit-aware features. Currently, FedPDFGuard relies on structural features derived from the object tree of each PDF file. While effective, these features primarily capture syntactic properties and may miss deeper semantic patterns or exploit behaviors. Future extensions of FedPDFGuard could incorporate high-level semantic features, such as JavaScript code analysis, embedded file characteristics, and API call patterns within streams. Additionally, exploit-specific signatures, such as suspicious /Launch actions, embedded executables, or obfuscated JavaScript, could help improve detection accuracy, especially against evasive threats.

2) Exploring advanced federated learning strategies. The current implementation of FedPDFGuard adopts a standard federated averaging protocol. However, more advanced FL methods, such as personalized federated learning [26], hierarchical FL [27], or adaptive aggregation techniques [28], could be employed to further enhance performance in heterogeneous data environments.

3) Addressing limitations and challenges. Although FedPDFGuard keeps raw data localized, FL is not immune to privacy attacks. Recent studies [29]–[31] have shown that model updates may leak information about training data. Incorporating differential privacy or secure aggregation protocols would strengthen defenses against such threats.

As our future work, we plan to extend FedPDFGuard by integrating deep learning models capable of automatically learning feature representations from raw PDF files. This could be achieved using graph neural networks (GNNs) [32] to process the object structure or transformer-based models [33] to parse embedded content. Additionally, we aim to incorporate on-device privacy-preserving mechanisms such as homomorphic encryption [34] or trusted execution environments [35]. Finally, we envision deploying FedPDFGuard in real-world environments to evaluate its effectiveness under operational constraints and dynamic threat landscapes.

VII. Conclusion

We presented FedPDFGuard, a novel federated learning-based framework for detecting PDF malware while preserving data privacy. In contrast to traditional centralized detec-

tion approaches, FedPDFGuard enables collaborative model training across distributed clients without sharing raw data. Our evaluation demonstrates that FedPDFGuard achieves a detection accuracy of 98.9% and a detection rate of 99.52%, performing competitively with centralized machine learning models such as CNN, SVM, and Random Forest. Looking ahead, extending FedPDFGuard to support raw PDF file processing and integrating advanced privacy-preserving techniques can further enhance its effectiveness and resilience against emerging threats.

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